

Free Tools To Make Your PCB SMARTER

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Simutron

Whether you want to run your Arduino firmware or debug it, Simutron provides an integrated development environment to meet your needs. This electronic circuit simulation freeware lets you design as well as simulate circuits. Internally, it uses a Simavr AVR processor simulator. You can use Simutron for 16MHz microcontroller units. In particular, it can be used in conjunction with its CAM front-end to develop CNC firmware without access to the real hardware.

SimulIDE

SimulIDE is a real-time circuit simulator freeware that supports microcontrollers such as PIC, AVR and even Arduino. Whether it is a small circuit or a circuit consisting of several nodes, SimulIDE makes design and simulation quick and easy for users. It comes with a limited set of features as it is intended for hobbyists and students who simply want to experiment with basic electronics projects.

Circuit Diagram

Quite useful for starters and students, this easy-to-use open source program lets users make electronic circuit diagrams and export them as images in SVG or PNG formats. They can

Some popular resources

Speccy. Using this software you can get details of every component in your computer, from the operating system of your PC to audio support, peripherals or network. The software supplements this information with temperatures of the components, so you can easily see if there's a problem!

FileZilla. It is a fast and reliable cross-platform FTP, FTPS and SFTP client. It has an intuitive graphical user interface with several important features.

IP Scanner. This open source freeware helps the master computer user to access other PCs in the network. One just needs a free Radmin viewer to operate a remote machine. Thus, all the systems in the network can be scanned for viruses using a friendly user interface system. The user can also remotely turn systems off. This software is designed for all levels of expertise.

Jaspersoft. You can create web-based reports using Jaspersoft tool. It picks information from various data sources and presents it in an easy-to-read, highly interactive format for business users.

design diagrams visually by placing components with the cursor. Component values can be modified using the properties editor.

TransistorAmp

Designers face issues in calculating transistor parameters like gain, amplification and biasing. What if they have a software that lets them calculate the perfor-

mance of transistors and design perfect schematics of common-base, common-emitter and common-collector amplifiers using combinations of transistors? This freeware tool does exactly that. It shows the values of other components required to get the desired results from a given input data.

Gputils

Microchip's PIC microcontroller is used in various development boards. Gputils is a collection of tools used by PIC microcontroller for serial programming and reprogramming in flash memory. Moreover, it is compatible with Microchip's tools MPASM, MPLINK and MPLIB. It runs on Linux, Windows as well as Mac OS platforms.

XYCE

It is an open source, SPICE-compatible analogue circuit simulator, capable of solving extremely large circuit problems by supporting large-scale parallel computing platforms. It also supports serial execution on all common desktop platforms, and small-scale parallel runs on Unix-like systems. It uses a differential-algebraic equation formulation. Being SPICE-compatible, it supports standard analysis methods. **EFY**

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